1. Prove that $(\Omega_A, \mathcal{F}_A, \mathbb{P}_A)$ and $(\Omega_B, \mathcal{F}_B, \mathbb{P}_B)$ are probability space.

(1) Model A: $\Omega_A = \{0, 1\}$, and it means it's discrete.

Thus, $\mathcal{F}_A = 2^{\Omega_A}$ is a power set of Ω_A. Thus, it is σ-algebra.

$\mathbb{P}_A(1) = 0.3$ and $\mathbb{P}_A(0) = 0.7$.

$\mathbb{P}_A(\Omega) = \mathbb{P}_A(1) + \mathbb{P}_A(0) = 1$, Thus, $\mathbb{P}_A$ is a probability measure.

(2) Model B: $\Omega_B = [0, 1]$ and it is continuous.

Thus, $\mathcal{F}_B = \mathcal{B}([0,1])$ is the σ-algebra for this sample space.

Since P_B is the uniform probability measure on $[0,1]$, $P_B([0,1]) = 1$.

Thus, P_B is a probability measure.

Thus, both model A and model B are probability spaces.

2. $F_X(x) = \mathbb{P}_A(X \leq x)$

$\mathbb{P}_A(X=0) = \mathbb{P}_A(0) = 0.7$

$\mathbb{P}_A(X=1) = \mathbb{P}_A(1) = 0.3$

$$\therefore F_X(x) = \begin{cases} 0, & x < 0 \\ 0.7, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

$F_Y(x) = \mathbb{P}_B(Y \leq x)$

$\mathbb{P}_B(Y=0) = \mathbb{P}_B(w > 0.3) = 0.7$

$\mathbb{P}_B(Y=1) = \mathbb{P}_B(w \leq 0.3) = 0.3$

$$\therefore F_X(x) = \begin{cases} 0, & x < 0 \\ 0.7, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

3. $\mathbb{P}_A$ and $\mathbb{P}_B$ are defined on different sample spaces.

$\mathbb{P}_A$ is defined on the sample space $\{0,1\}$, while $\mathbb{P}_B$ is defined on $[0,1]$.

The distribution function is defined on X and Y seperately.

They show the probability when the X or Y equals certain number. In this case, they are same.

4. You can't determine which model generated the data because both models satisfy $\mathbb{P}(Z=1) = 0.3$ and $\mathbb{P}(Z=0) = 0.7$.

5.

	A	B
Randomness	The set is discrete. The randomness lies in which number we choose. The probability lies in for each number.	The set is continuous. The randomness lies in a range. The probability lies in which range the number is.

6. $\because B \cup C = \{011, 111, 101, 010, 110, 001\}$ ~~$\{101, 100, 110, 000, 001, 010\}$~~

$\because B^c = \{000, 001, 100, 101\}$

$C^c = \{000, 010, 100, 110\}$

$\therefore B^c \cap C^c = \{000, 100\}$

$\Rightarrow \therefore (B \cup C)^c = \{000, 100\}$

$\therefore B^c \cap C^c = (B \cup C)^c$

$\because (A \cup B) = \{010, 011, 100, 101, 110, 111\}$

$\therefore (A \cup B) \cap C = \{011, 101, 111\}$

$\because A \cap C = \{101, 111\}$, $B \cap C = \{011, 111\}$

$\therefore (A \cap C) \cup (B \cap C) = \{011, 101, 111\}$

$\therefore (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

LLM

I'll model the LLM that guesses the next word.

Ω is the set of all possible words.

σ-algebra is 2^{Ω}.

The probability is $\frac{\#\text{occurance in training data}}{\#\text{ all words' occurance}}$